

Extended Reality Based Job Readiness Training System for Neurodiverse Population

Jingyi (Jenny) Huang, Sanjay Goel, Reza Jari
FACETS, Massry School of Business, University at Albany, SUNY



The Employment Gap

15-20%

of people worldwide identify as neurodivergent

85-90%

unemployment and underemployment rate

- Neurodivergent employees bring new perspectives and creative thinking, and employers who recruit them report gains in **innovation and productivity**.
- Stable jobs not only enable **independent living**, but also give a **sense of purpose, meaning, and identity** for neurodiverse people.
- Yet the population faces severe labor market exclusion.

Why Current Systems Fail

- Support drops off.** Formal support largely ends when secondary education ends, and transition services lack standardized and long-term focus.
- Outdated practice.** Many career counselors rely on old frameworks, so few people are trained in vocational skills for competitive employment.
- Too little experience.** Internships and work placements are scarce, especially for autistic people.
- Biased assessment.** Interviews reward social polish and impression management. Standard tests were built for neurotypical people, and AI CV screeners penalize atypical education or work histories.

Different Neurodiverse Needs

- Autism (ASD).** Differences in social communication and executive function, plus sensory hyper- or hyposensitivity. The “double empathy problem” means misunderstanding runs both ways.Masking to fit in can lead to burnout.
- ADHD.** Differences in attention, inhibitory control, and planning, time management and multitasking can be hard. Hyperfocus is a strength that can tip into overwork and burnout.
- Intellectual disability (ID).** Internships and work placements are scarce, Limits in reasoning, learning and adaptive behavior. It cooccurs with autism in 30-50% of cases. This group has the highest labor market exclusion.

Why Extended Reality

Extended reality (XR) spans virtual, augmented, and mixed reality. Combined with AI, IoT, and 5G, it can now deliver richer, real-time, personalized experiences. Compared with classroom, video, or on-the-job training, XR offers:

- Faster and stickier learning.** Higher motivation, retention, accuracy, and confidence.
- Low stakes practice.** A realistic, danger-free, stress-controlled environment with no consequences attached to failure.
- Sensory control.** Remove distractions and background noise for people with sensory sensitivities.
- Individualization.** Tailor tasks and scenarios to each trainee based on goals, needs and abilities.
- Scale.** More flexible and cost-effective than one-on-one coaching.

The Research Gap

Evidence shows support for the effectiveness of XR for job tasks and soft skills, but its use with neurodivergent people is underexplored. Most programs are standardized, with limited customization and cannot meet the diversity of needs in this population. Little exists that prepares people for different types and levels of jobs.

Our Project

Harness AI and XR to improve employment readiness by improving both training delivery and skills assessment.

- Assess.** Use AI to build an assessment tool that measures skills and abilities more appropriately.
- Train.** Develop an adaptive training system matched to each user’s target job, adjusted according to needs and abilities.

Three Research Focus Areas

Social Interaction

- VR and AR social skills training improves emotion recognition, turn-taking, and conversational flow in autistic people, but is hard to tailor to varied preferences. We will use AI to read users’ cues and build adaptive, personalized sequences.

Stress Management

- Autistic adults report higher perceived stress, linked to lower quality of life. Virtual beaches and forests quickly reduce negative emotions. Wearable haptics that deliver calming pressure of rhythm can add sensory support, and generative AI can personalize it.

Job training

- Comprehensive, job-specific training is missing. Beyond interviews, we will cover teamwork, communication, task management, and problem-solving across job types and levels, with AI-informed assessment of what each person can actually do.

Selected References

Bölte, S. et al. (2025). Career guidance and employment issues for neurodivergent individuals: a scoping review and stakeholder consultation. *Human Resource Management*, 64(1), 201-227.

Davies, J. et al. (2023). Access to employment: A comparison of autistic, neurodivergent and neurotypical adults’ experiences of hiring processes in the United Kingdom. *Autism*, 27(6), 1746-1763.

Doyle, N. (2020). Neurodiversity at Work: A Biopsychosocial Model and the Impact on Working Adults. *British Medical Bulletin* 135: 108–125.

Eckert, D, & Mower, A. (2020). The effectiveness of virtual reality soft skills training in the enterprise: a study [Internet]. London: PricewaterhouseCoopers. Available from: <https://www.pwc.com/us/vlearning>.

Ezerins, M. E. et al.(2024). Autism and employment: A review of the “new frontier” of diversity research. *Journal of Management*, 50(3), 1102–1144.

Howard, M. C., & Gutworth, M. B. (2020). A meta-analysis of virtual reality training programs for social skill development. *Computers & Education*, 144, 103707.

Makransky, G. et al. (2021). Immersive virtual reality increases liking but not learning with a science simulation and generative learning strategies promote learning in immersive virtual reality. *Journal of Educational Psychology*, 113(4), 719–735.

Roux, A. M. et al. (2015). *National autism indicators report: Transition into young adulthood*. Life Course Outcomes Research Program’, AJ Drexel Autism Institute, Drexel University, Philadelphia, PA.

Seaman, R. L., & Cannella-Malone, H. I. (2016). Vocational skills interventions for adults with autism spectrum disorder: A review of the literature. *Journal of Developmental and Physical Disabilities*, 28(3), 479–494.

Tao, T., & Fu, W. (2025). The effectiveness of extended reality technology interventions on vocational skills for individuals with autism and intellectual disabilities: A meta-analysis. *Journal of Autism and Developmental Disorders*.

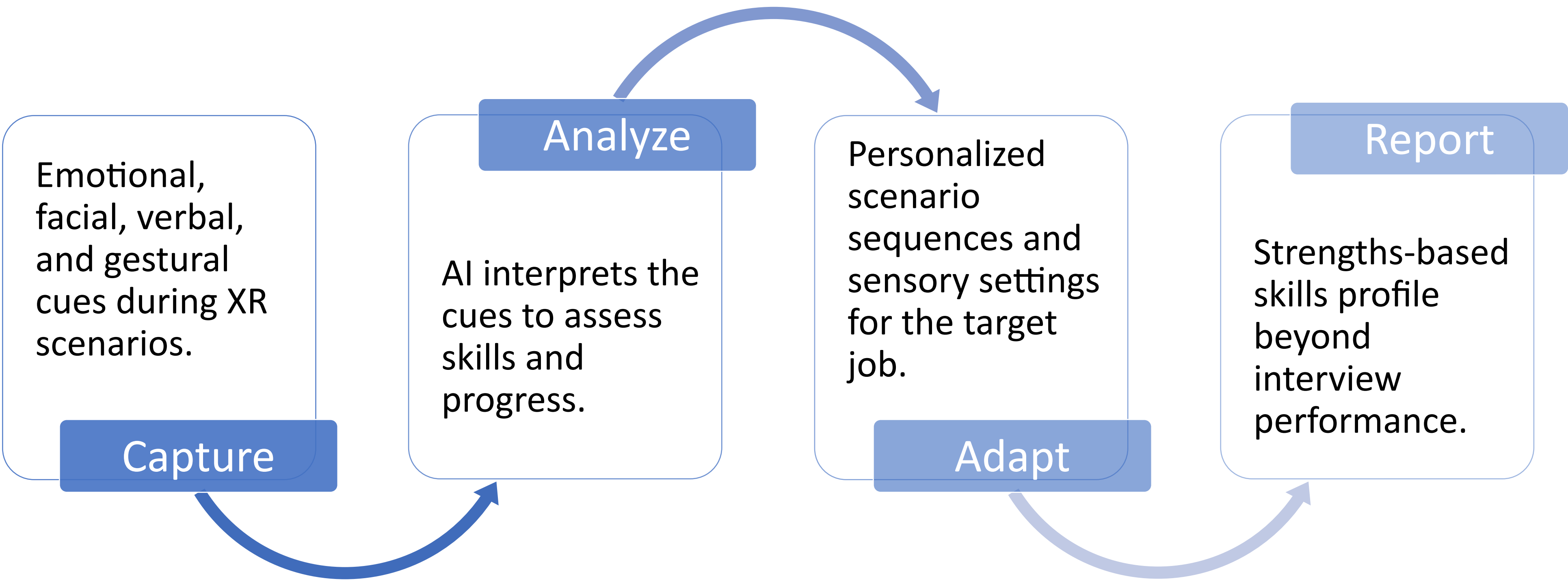


Fig1. How the system adapts